Advanced ABM
Assignment 3

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Problem 1

The Party's size grew at a positive rate proportional to its size, but every day k students dropped out.

a) Formalize this narrative using the Construction Kit.

b) Compute any equilibria algebraically.

c) Plot the right-hand side of your model (its $G(x)$) and graphically—using flow arrows—determine the stability of any equilibria identified in part (b).

Answer 1:

- We can recognize the size of the party to be denoted by P and the change in the party's size to be a function of time, given by $G(x)$. Then, we can formalize this narrative using the following difference equations and its continuous analogue:

$$\frac{dP}{dt} = \alpha P - k \mid \alpha \in \mathbb{R}_{\geq 0}, k \in \mathbb{N}$$

$$P_{t+1} = \alpha P_t - k \mid \alpha \in \mathbb{R}_{\geq 0}, k \in \mathbb{N}$$

$$\implies G(x) = \alpha P - k \mid \alpha \in \mathbb{R}_{\geq 0}, k \in \mathbb{N}$$

- The equilibria can then be found by setting the growth rate equal to zero, such that:

$$\begin{aligned} G(x) = 0 \implies 0 &= \alpha P - k \\ k &= \alpha P \\ P &= \frac{k}{\alpha} \\ \implies P^* &= \frac{k}{\alpha} \end{aligned} \tag{Party Equilibria}$$

Then, we can analyze the stability of the equilibria P^* which is given by $G'(P) = \frac{d}{dP}(\alpha P - k) = \alpha$ and since $\alpha \in \mathbb{R}_{\geq 0}$, we know that the equilibria associated with $P^* = \alpha$ must be unstable as $\alpha > 0 \implies G(P^*) > 0$.

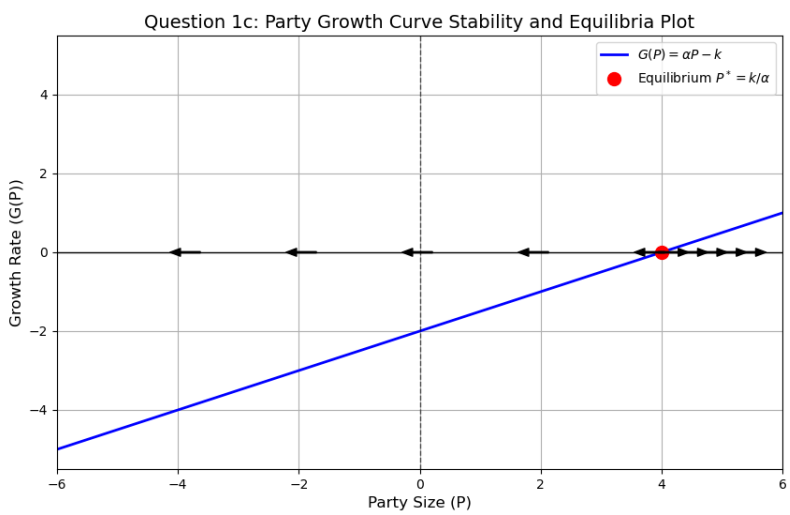

Figure 1: Party Stability and Equilibria Plot

Problem 2

A deadly new flu variant has an $R_0 \approx 6$. There is a safe and perfectly effective vaccine. If administered at the outset of the epidemic (when the susceptible fraction is essentially 1):

a) What level of vaccine coverage is required for herd immunity?

b) Assuming that this vaccination program had been fully implemented, would it be responsible for the government to announce: "Herd immunity has been achieved, so the epidemic is essentially over."

Answer 2:

- The level of vaccine coverage required for herd immunity when $R_0 = 6$ can be described through an SIR model and the equilbria of the system when the fraction of susceptibles in the model is nearly 1, such that:

$$\begin{aligned} \text{Herd Immunity} &= 1 - \frac{1}{R_0} \\ &= 1 - \frac{1}{6} \\ &= 0.833 \\ &\implies 83\% \text{ of the population must be vaccinated to achieve herd immunity} \end{aligned}$$

- Herd immunity does reduce the rates of transmission but doesn't eliminate a disease or the chance of an outbreak entirely, especially in localized communities that are specifically intervention/vaccine-averse. Herd immunity requires $((1 - \frac{1}{R_0}) \cdot 100)$ percent of the population to be vaccinated against the disease so it cannot proliferate as easily, but if the entire population were to homogeneously interact, vaccinated individuals could potentially spread the disease to the vaccine-free individuals.

Problem 3

In the Civil Violence model, let all variables (hardship, legitimacy, grievance, cops, etc.) be fixed. Unlike the original model, however, now cops are indiscriminate, and are known to be indiscriminate, arresting all agents within their vision whether rebellious or not. The cops, in this sense, are more aggressive. Does this bolster or undermine deterrence?

Answer 3:

- If cops arrest all agents indiscriminately, the perceived risk of rebelling increases for the population. This higher risk strengthens rates of deterrence, and so, fewer individuals are likely to rebel. Despite that though, increasingly aggressive policing could also generate resentment, which may potentially increase tension over time, which could eventually lead to higher rates of rebellion.

Problem 4

Open the NetLogo program named `2023PursuitLanchesterAgentMelee`. Modify the `attack-and-occupy` procedure so that, after killing their target, the agent clones (using the `hatch` command) an identical agent with a hatch-probability between zero and one. Add a slider to the interface allowing the user to adjust the hatch-probability.

a) Submit all code and a snapshot of the new interface.

b) Set `blue-accuracy = red-accuracy = 0.015` and `kill-radius = 3`. Run the model a few times at hatch-probabilities of 0.0 and 1.0 and compare and discuss the dynamics and equilibrium spatial patterns.

Answer 4:

- Snapshot of the new interface:

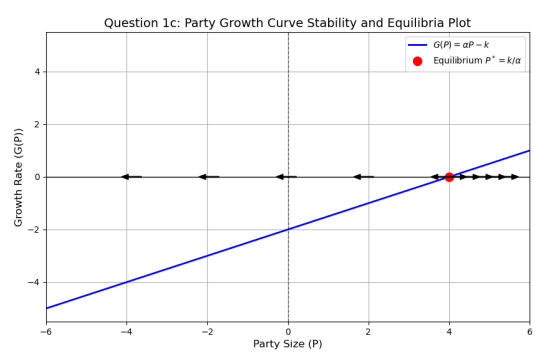

Figure 2: Updated Interface

- As the hatch probability approaches 1, the difference between the final population of the red and blue turtles deviates more and more from each other, and the spatial distribution of agents tends to become more and more random.

Problem 5

In combat between Red and Blue forces, each side's rate of attrition (daily loss rate or derivative) is proportional to the opponent's force size, but every day Red gets fixed reinforcements of X_R units and Blue gets daily reinforcements of X_B units.

a) Mathematize this narrative.

b) Derive the equilibrium values of R and B.

c) Open the Lanchester Theory NetLogo model. Leaving all settings as shown, modify the `goSquare` code, adding to the interface one slider for the red daily reinforcements (X_R) and another for the daily Blue reinforcements (X_B), each ranging from 0 to 50. With both set to zero, we get mutual annihilation. Confirm that with $X_R = 90$ and $X_B = 45$ we get the predicted positive equilibrium values, and show a plot of the dynamics.

Answer 5:

- The system of differential equations that describes the attrition rate of the Red and Blue forces is given by:

$$G_B(t) = X_B - \alpha R \qquad \text{(Blue Force Growth Function)}$$
$$G_R(t) = X_R - \beta B \qquad \text{(Red Force Growth Function)}$$

- The equilibrium of R and B can be found by setting the equations $G_B(t) = 0$ and $G_R(t) = 0$ such that:

$$\begin{aligned} G_B(t) = 0 \implies X_B - \alpha R &= 0 \\ \alpha R &= X_B \\ R &= \frac{X_B}{\alpha} \\ \implies R^* &= \frac{X_B}{\alpha} \qquad \text{(Blue Force Equilibria)} \end{aligned}$$

$$\begin{aligned} G_R(t) = 0 \implies X_R - \beta B &= 0 \\ \beta B &= X_R \\ B &= \frac{X_R}{\beta} \\ \implies B^* &= \frac{X_R}{\beta} \qquad \text{(Red Force Equilibria)} \end{aligned}$$

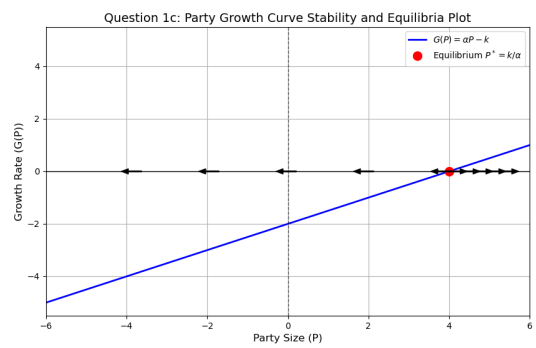

Figure 3: Reinforcement Modification Dynamics

Problem 6

You may consider the following dynamical system in either of the following (continuous or discrete) ways:

$$\frac{dx}{dt} = ax - bx^2$$

or

$$x_{t+1} - x_t = ax_t - bx_t^2$$

where the parameters a and b are positive and $x \geq 0$. The right-hand sides are the rates of change we've been denoting by $G(x)$.

a) Derive algebraic expressions for any and all equilibria (values of x where there is zero growth).

b) What are the equilibria if $a = 10$ and $b = 2$?

c) For these values, plot the growth curve $G(x)$.

d) Classify the stability of any equilibria using arrow flow lines (phase portraits).

e) Compose an English-language narrative—a story—that would be modeled or mathematized by the given dynamical equation. The story need not be about humans or even living creatures. Be prepared to share your story with the class.

Answer 6:

- The equilbria can be found by setting the change between $t+k$ and t equal to zero and solving such that:

$$\begin{aligned} \frac{dx}{dt} = 0 \implies ax - bx^2 &= 0 \\ bx^2 &= ax \\ x &= \frac{a}{b} \\ \\ \implies x^* &= \frac{a}{b} : a, b \in \mathbb{R}_{>0}, x \in \mathbb{R}_{\geq 0} && \text{(Non-trivial Equilibria)} \\ x^* &= 0 && \text{(Trivial Equilibria)} \end{aligned}$$

- The equilibria if $a = 10$ and $b = 2$ is $x^* = \frac{10}{2} = 5$, and since $G'(5) = 10 - 2(2 \cdot 5) = -10 < 0$, the non-trivial equilibria $x^* = \frac{a}{b}$ is unstable. The trivial equilibria $x^* = 0$ gives us $G'(0) = 10 - 2(2 \cdot 0) = 10 > 0$, which means the equilibria associated with $x^* = 0$ is unstable.

- The first equilibria is the trivial equilibria that occurs when $x^* = 0$ and is unstable, as the flow lines travel away from that specific critical point. The second equilibria occurs when $x^* = \frac{a}{b}$ and must be stable, since the quadratic will only have two roots, and by the Mean Value Theorem, the equilibria must alternate for a well-defined growth function.

- Let x_t represent the number of flowers in a meadow. Each flower reproduces at a rate proportional to the current population ax, while competition for limited space and nutrients causes deaths at a rate proportional to the square of the population bx^2. Over time, the population increases when the population is small, and the growth slows as resources become increasingly scarce, ultimately stabilizing when the population of flowers reaches the environment's carrying capacity.

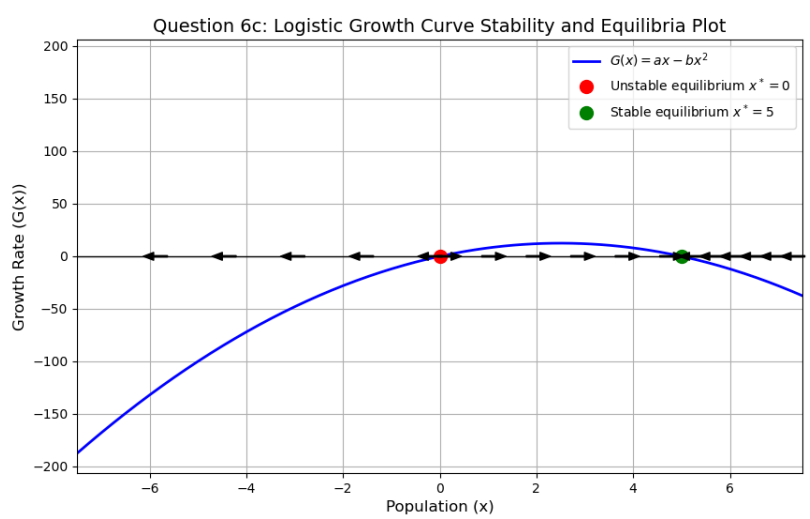

Figure 4: Flower Population Stability and Equilibria Plot